

HARSH RAI

Full Stack Developer (MERN) | Backend Engineer | B.Tech CSE 26 - FGIET, Raebareli
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SUMMARY

Sole developer and backend-focused MERN Stack engineer, graduating B.Tech CSE (FGIET, 2026). Built CodeArena - a LeetCode-style contest platform - end-to-end: MongoDB schema design, Express middleware pipeline, REST APIs, JWT/RBAC auth, AI integration, and a multi-role React frontend. Seeking SDE Intern, Full Stack, Backend Developer, and SDE-1 roles at product-focused engineering teams.

TECHNICAL SKILLS

Languages	JavaScript (ES6+), Java, HTML5, CSS3
Frontend	React.js, React Hooks, Tailwind CSS, React Router, Axios, Monaco Editor
Backend	Node.js, Express.js, RESTful API Design, Middleware Architecture, MVC
Database	MongoDB, Mongoose ODM, Schema & Index Design
Auth & Security	JWT, RBAC, OTP Email Auth (Nodemailer), Bcrypt, Input Sanitization, OWASP Top 10
AI & Tools	OpenRouter API, Prompt Engineering, Git, GitHub, Postman, VS Code
CS Fundamentals	Data Structures & Algorithms, OOP, SDLC, Agile Basics

PROJECTS

CodeArena - Real-Time Coding & Contest Platform | Sole Developer

React | Node.js | Express | MongoDB |
JWT | Tailwind | OpenRouter API

<https://github.com/raiharsh24/finalproject>

- Designed full system architecture - MongoDB schemas, Express middleware pipeline, RESTful routing - for a three-role platform (Student / Teacher / Admin) with 20+ typed API endpoints and centralized error handling.
- Engineered a stateful contest engine: countdown timer, live leaderboard with freeze mechanics, submission locking, and weekly scheduling - replicating competitive programming platform behavior at the application layer.
- Implemented multi-layer authentication: JWT access tokens, OTP email verification (Nodemailer), Bcrypt hashing, and RBAC middleware enforcing least-privilege access across all protected routes and admin operations.
- Integrated OpenRouter API as AI mentor - prompt-engineered for context-aware hint generation, live debugging assistance, and a floating coding assistant active throughout contest and practice sessions.
- Shipped SaaS-style responsive UI (Tailwind CSS + Monaco Editor) with admin analytics dashboard, problem bank, quiz management, and contest moderation tools - full platform governance in one interface.

Sakshi Hardware - Full Stack Business Website

React | Node.js | Express | MongoDB

<https://github.com/raiharsh24/sakshi-hardware>

- Architected a product catalog platform with category navigation, customer inquiry management, and Mongoose-backed CRUD APIs; applied modular Express routing and input validation middleware for data integrity.
- Applied MVC separation and schema-driven data modeling to ensure maintainability and clean backend-frontend contract boundaries.

Basic Banking System | Budget Tracker

Java | OOP

<https://github.com/raiharsh24/personal-finance-tracker>

- Console-based financial applications demonstrating OOP design - encapsulation, modular class structure, and validated transaction logic - for account management and budget tracking.

EDUCATION

B.Tech - Computer Science Engineering, FGIET, Raebareli, UP | CGPA: 7.5+ (Expected)

May 2026

Class XII & Class X - Nagaji Saraswati Vidya Mandir Senior Secondary School

CERTIFICATIONS

Cybersecurity Training - TryHackMe | Web Application Hacking, Network Security, Linux Privilege Escalation, OWASP Top 10